

The Hierarchical Observer, Step by Step

Every parameter, and how each one combines into the next

Neuromatch Bayesian Modelling project

The figure below is the whole model; each arrow carries the equation for that step. The two sections after it (a) define every parameter and (b) walk through the arrows one at a time, stating for each step its **inputs** → **operation** → **output** and the logic of how the quantities combine.

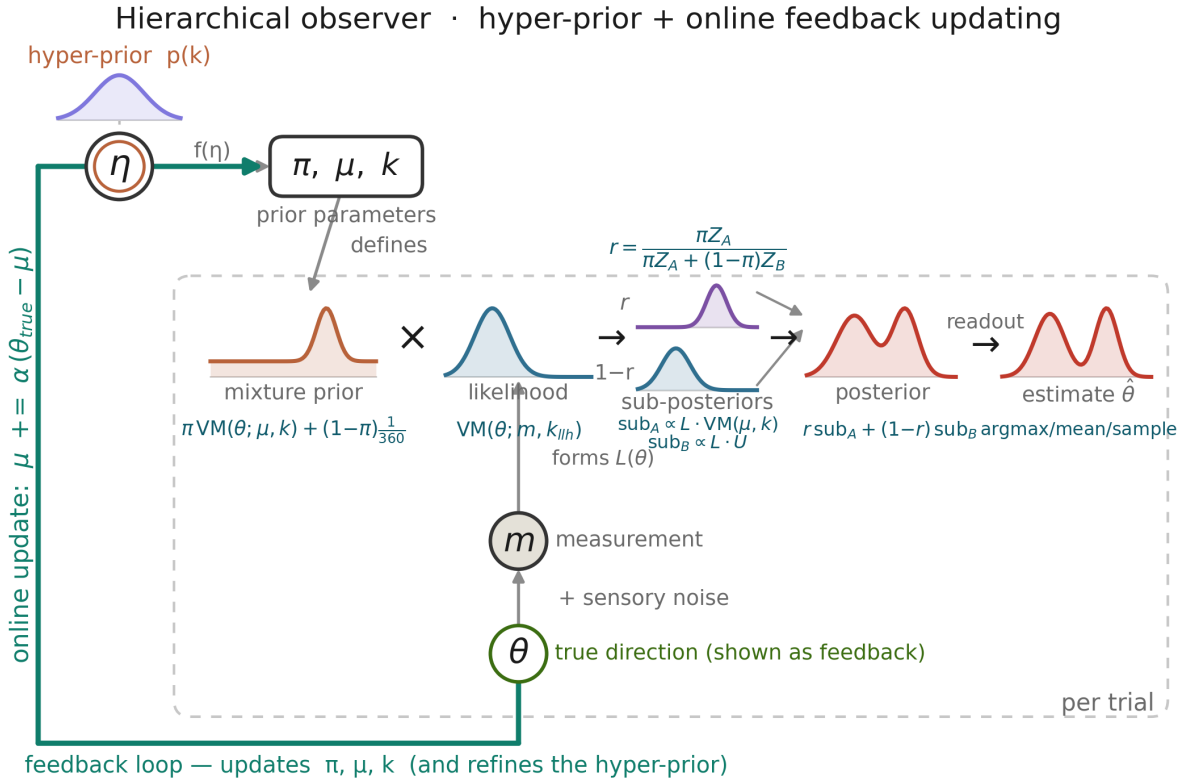


Figure 1: Hierarchical observer with the hyper-prior and the online feedback loop. Each transition is labelled with the operation that combines its inputs.

Parameter dictionary

Grouped by role: the *hyperparameters* (learned slowly), the *prior parameters* they produce, the *per-trial* quantities, and the *fixed nuisance* parameters.

Symbol	What it is	How it is set / its role
η	The hyper-prior variable	Belief about the prior itself; the top-level knob that produces (π, μ, k) .
$p(\eta)$	The hyper-prior	A distribution over η — a prior over the parameters of the prior.
$f(\eta)$	The link map	Turns η into the concrete prior parameters (π, μ, k) .
π	Mixing weight	$P(z = \text{peaked}) \in [0, 1]$: how much of the prior is the peaked component vs the uniform floor.
μ	Prior mean/mode	The centre direction the prior favours — 225° in the task [1].
k	Prior concentration	Prior <i>strength</i> ; larger k = narrower prior ($\sigma \approx 1/\sqrt{k}$).
θ	True direction	The stimulus direction on the trial; also shown as feedback afterward.
m	Measurement	The noisy sensory reading of θ .
k_{lh}	Sensory reliability	Concentration of the measurement noise; grows with motion coherence.
$L(\theta)$	Likelihood	$\text{VM}(\theta; m, k_{lh})$ — what the measurement says about direction.
$A(\theta)$	Peaked prior component	$\text{VM}(\theta; \mu, k)$.
$B(\theta)$	Uniform prior component	$U = 1/360$ (flat).
Z_A, Z_B	Model evidences	$Z_A = \sum_{\theta} L A$, $Z_B = \sum_{\theta} L B = 1/360$: how well each cause predicts m .
r	Responsibility	$P(z = \text{peaked} \mid m)$: the trial-specific weight on the peaked cause.

Symbol	What it is	How it is set / its role
$\hat{\theta}$	Estimate	The reported response after readout, motor noise, and lapse.
k_{motor}	Motor precision	Concentration of the response (motor) noise.
p_{rand}	Lapse rate	Probability of a random guess on a trial.
α	Learning rate	Step size of the online update of μ (and, analogously, k, π).

How the quantities combine, step by step

Each step is one arrow in the figure.

Step 1 — hyper-prior to prior parameters ($\eta \rightarrow \pi, \mu, k$). Inputs: η . Operation: $(\pi, \mu, k) = f(\eta)$. Output: the three prior parameters. *Logic:* η is your belief about what the prior looks like; f reads off its current best values for the mixing weight, centre, and strength. If you are uncertain about η , you instead carry the whole distribution $p(\eta)$ and average over it later (Step 2, predictive form).

Step 2 — prior parameters to the prior ($\pi, \mu, k \rightarrow p(\theta)$). Inputs: π, μ, k . Operation:

$$p(\theta) = \pi \text{VM}(\theta; \mu, k) + (1 - \pi) \frac{1}{360}.$$

Output: the mixture prior over direction. *How they combine:* μ places the bump, k sets its width, and π mixes it with a uniform floor — π is the height of the bump relative to the flat baseline. With uncertainty over η , replace this by the predictive prior $\int p(\theta | \eta) p(\eta) d\eta$, which is wider and heavier-tailed [3].

Step 3 — true direction to measurement ($\theta \rightarrow m$). Inputs: θ, k_{lh} . Operation: $m \sim \text{VM}(\theta, k_{lh})$. Output: a noisy measurement. *Logic:* the sensory system corrupts the true direction; k_{lh} (reliability) controls how tightly m scatters around θ .

Step 4 — measurement to likelihood ($m \rightarrow L$). Inputs: m, k_{lh} . Operation: $L(\theta) = \text{VM}(\theta; m, k_{lh})$. Output: the likelihood over all directions. *Logic:* the same von Mises, now read as a function of the hypothesised direction θ given the observed m .

Step 5 — prior $\times$ likelihood to the two sub-posteriors. Inputs: $L, A = \text{VM}(\mu, k), B = U$. Operation:

$$\text{sub}_A \propto L \cdot A, \quad \text{sub}_B \propto L \cdot B, \quad Z_A = \sum_{\theta} LA, \quad Z_B = \sum_{\theta} LB = \frac{1}{360}.$$

Output: two normalised sub-posteriors and their evidences. *How they combine:* because the prior is a **sum** of two parts and Bayes **multiplies**, the product splits — one posterior per component (multiplication distributes over addition). sub_A is the prior-pulled

estimate; sub_B is just the likelihood. Z_A, Z_B measure how much total probability each component gives the data.

Step 6 — evidences to responsibility ($Z_A, Z_B \rightarrow r$). Inputs: π, Z_A, Z_B . Operation:

$$r = \frac{\pi Z_A}{\pi Z_A + (1 - \pi) Z_B}.$$

Output: the responsibility $r \in [0, 1]$. *How they combine*: this is Bayes one level up — the prior odds $\pi/(1 - \pi)$ times the evidence ratio Z_A/Z_B , renormalised. r is large when the measurement lands where the peaked prior predicts it (near μ), small when it lands far away.

Step 7 — sub-posteriors and responsibility to the posterior. Inputs: $\text{sub}_A, \text{sub}_B, r$. Operation:

$$p(\theta | m) = r \text{sub}_A + (1 - r) \text{sub}_B.$$

Output: the (generally bimodal) posterior. *How they combine*: a convex blend — r slides mass between the prior-pulled mode and the likelihood mode. Equivalently, the sum rule $p(\theta | m) = \sum_z p(z | m) p(\theta | m, z)$ [2].

Step 8 — posterior to estimate ($p(\theta | m) \rightarrow \hat{\theta}$). Inputs: the posterior; $k_{\text{motor}}, p_{\text{rand}}$. Operation: pick a readout $g \in \{\text{mean, argmax, sample}\}$, then blur and lapse:

$$\hat{\theta} = [(1 - p_{\text{rand}}) g(p(\theta | m)) + p_{\text{rand}} U] \circledast \text{VM}(0, k_{\text{motor}}).$$

Output: the reported estimate. *How they combine*: the readout collapses the posterior to a value (averaging $\rightarrow$ one mode; argmax/sample $\rightarrow$ two modes); $\circledast$ is circular convolution with motor noise; p_{rand} mixes in a uniform guess.

Step 9 — feedback to the prior parameters ($\theta_{\text{true}} \rightarrow \pi, \mu, k$). Inputs: the fed-back true direction θ_{true} ; learning rate α . Operation (online form):

$$\mu \leftarrow \mu + \alpha(\theta_{\text{true}} - \mu), \quad k, \pi \text{ updated toward the observed statistics;}$$

the exact form is Bayes on the hyperparameters, $p(\eta | \theta_{1:t}^{\text{true}}) \propto p(\eta) \prod_s p(\theta_s^{\text{true}} | \eta)$. Output: updated (π, μ, k) (and a refined $p(\eta)$) for the next trial. *How they combine*: the loop closes here — the true directions shown as feedback are samples of the prior, so learning the prior is estimating a distribution from those samples; the delta rule with rate α is the online, point-estimate version [23, 24].

References

- [1] Laquitaine, S. & Gardner, J.L. (2018). *Neuron* 97, 462–474. [2] Bishop, C.M. (2006). *Pattern Recognition and Machine Learning*, ch. 9. [3] Gelman, A. et al. (2013). *Bayesian Data Analysis*, 3rd ed., ch. 5. [23] Nassar, M.R. et al. (2010). *J. Neurosci.* 30, 12366–12378. [24] Wilson, R.C., Nassar, M.R. & Gold, J.I. (2013). *PLoS Comput. Biol.* 9, e1003150.